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QUANTITATIVE PORTFOLIO OPTIMIZATION AND STYLE ANALYSIS

INTRODUCTION

This project aims to build, test and find the best investment portfolio for clients from using ten highly liquid equities which are TSLA, WMT, BAC, GS, LLY, MRK, GOOG, META, AAPL and XOM. The base period of the analysis covers both in-sample estimation (September 1, 2023 — September 30, 2025) and out-of-sample testing (October 1, 2025 — December 31, 2025). This report presents a long-term asset allocation methodology that combines mathematical rigor with practical client constraints by utilizing Modern Portfolio Theory (MPT), factor exposure models, Monte Carlo simulations and the Black-Litterman framework.

PART 1: STANDARD MEAN-VARIANCE PORTFOLIO OPTIMIZATION

1.1 METHODOLOGY AND MATHEMATICAL FRAMEWORK

An asset allocation strategy is built on the MVO framework developed by Markowitz (1952). The aim is to do improved risk adjusted return of the portfolio quantified by Sharpe ratio, under various constraints.

We define the optimization problem as maximizing the objective function:

$$\text{Maximize } S_p = \frac{\mu_p - R_f}{\sigma_p}$$

Here μ_p is the expected portfolio return; R_f is the risk-free rate (assumed at 4% annualized); and σ_p is the portfolio volatility. We can write the portfolio return and variance as follows:

$$\mu_p = w^T \mu$$

$$\sigma_p^2 = w^T \Sigma w$$

In this case, w is the vector of asset weights, μ is a vector of historical mean returns, and Σ is the variance-covariance matrix of the asset returns.

1.2 IN-SAMPLE OPTIMIZATION RESULTS (STEP 1 AND STEP 2)

The daily return of both portfolios is constructed using the data from in-sample period.

Portfolio 1 (Unconstrained): The only constraints applied were that the weights must sum to 1 ($\sum w_i = 1$) and no short selling is allowed ($w_i \geq 0$). Typical of unconstrained MVO, the optimizer concentrated capital to a small selection of assets that had the best backtested risk-adjusted results – in this case AAPL and GS received weighted allocations while six out of the ten assets got zero allocation.

Portfolio 2 (Constrained): As a result of some clients confirming their own risk-management preferences, a maximum position limit was put in place. As the client wanted a cap of 15%, we defined an aggressive rule, with weights to be set at maximums of 18% per asset ($0 \leq w_i \leq 0.18$). This made the optimizer spread out and map capital more broadly across the asset universe.

In-Sample Portfolio Allocations (Training Period)

Step 1: Optimal Weights (Unconstrained)

Asset	Weight
TSLA	0.0000
WMT	0.0000
BAC	0.0523
GS	0.3550
LLY	0.0000
MRK	0.1450
GOOG	0.0000
META	0.0000
AAPL	0.4477
XOM	0.0000

Metric	Value
Expected Return	0.4218
Volatility	0.1908
Sharpe Ratio	2.0007

Step 2: Constrained Weights (Max 18%)

Asset	Weight
TSLA	0.0000
WMT	0.1800
BAC	0.1800

GS	0.1800
LLY	0.0883
MRK	0.1800
GOOG	0.0000
META	0.0000
AAPL	0.1800
XOM	0.0117

Metric	Value
Expected Return	0.3826
Volatility	0.1896
Sharpe Ratio	1.8064

And as expected mathematically, when the 18% box constraint was added, in-sample theoretical Sharpe ratio reduced slightly from 2.0007 to 1.8064 because the optimizer is restrained from fully taking advantage of historical mean returns of best assets (Bodie, Kane, & Marcus, 2014).

1.3 EFFICIENT FRONTIER ANALYSIS (STEP 3)

In order to illustrate the effect of the constraints, both scenarios (unconstrained and constrained) were plotted in terms of their efficient frontiers.

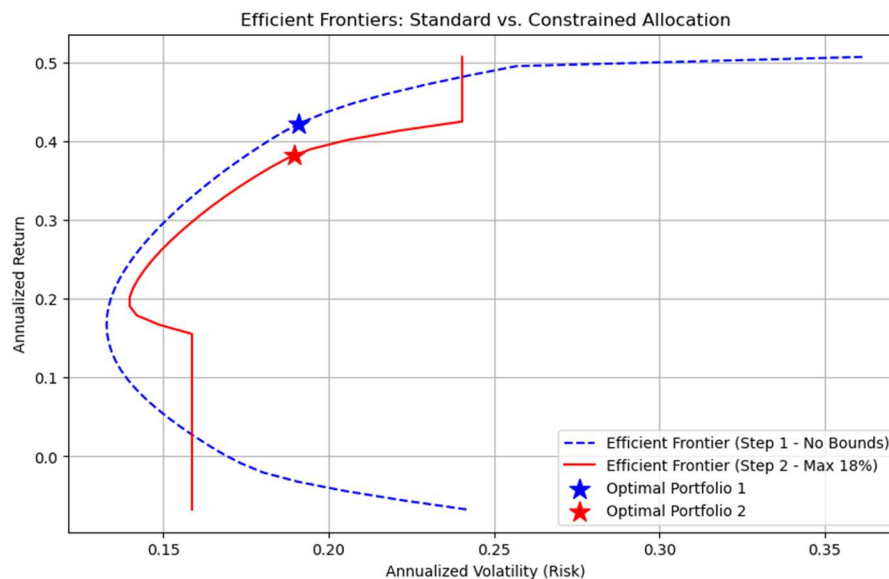


Figure 1: Efficient Frontiers comparing the unconstrained asset allocation (blue dashed) with the 18% maximum weight constrained allocation (red solid).

Figure 1 The basic tradeoff in constrained optimization. The Y axis gives the objective, and the X axis gives the constraint. We see that the unconstrained frontier (blue) always dominates the constrained frontier (red) over all volatility levels in the in-sample data. The constrained frontier is bounded upwards, as no asset can take up more than an 18% weight, and therefore the portfolio cannot access the extreme high

returns (and volatility) associated with a fully invested portfolio comprised of the best performing historical asset over that period.

1.4 OUT-OF-SAMPLE PERFORMANCE EVALUATION (STEP 4)

Although Portfolio 1 had much better in-sample metrics, it was extremely concentrated with a big risk to overfitting the noise of past history (De Prado, 2018). To check for robustness, both portfolios were kept unchanged and tested on a held-out out-of-sample period of October 1, 2025, to December 31, 2025.

Out-of-Sample Performance (Oct 1, 2025 – Dec 31, 2025)

Metric	Step 1: Unconstrained	Step 2: Max 18% Constraint
Cumulative Return	9.3033%	12.0134%
Annualized Sharpe Ratio	2.2493	3.1054
Maximum Drawdown	-4.6895%	-2.9200%

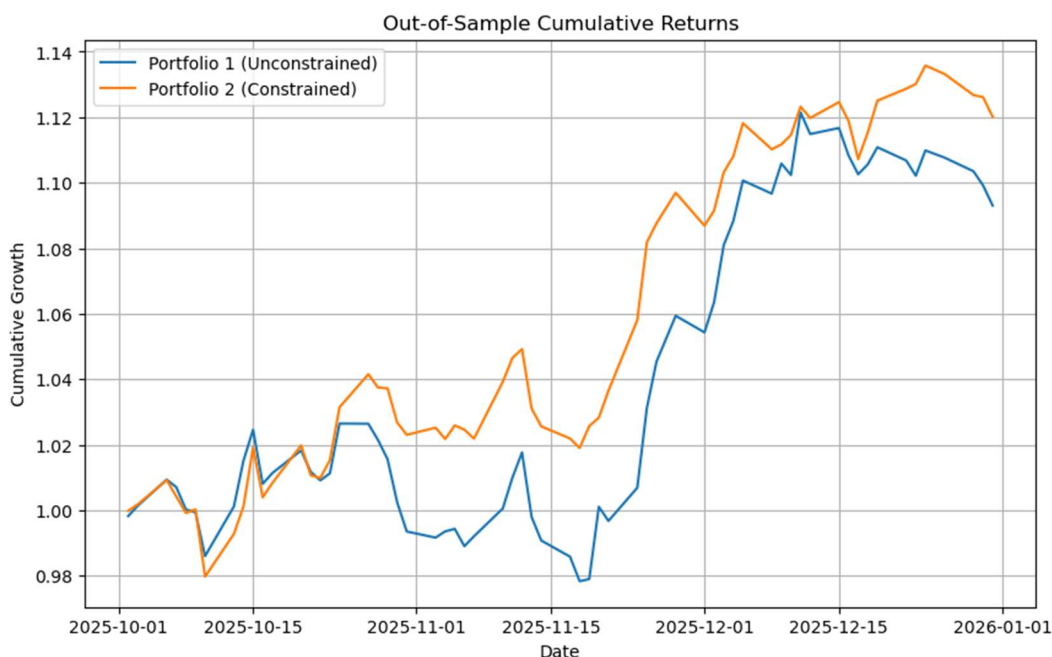


Figure 2: Out-of-sample cumulative growth of Portfolio 1 vs. Portfolio 2.

In-sample projections contrast sharply with out-of-sample results. Cumulative return was 12.01% for Portfolio 2 (Constrained) and 9.30% for Portfolio 1, while the out-of-sample Sharpe ratio was significantly in favor of Portfolio 2: 3.1054 vs. 2.2493. In addition, the Maximum Drawdown of Portfolio 2 was -2.92%, while Portfolio 1 showed a Maximum Drawdown of -4.68%, which provides an additional demonstration of better downside protection for Growth by Sector. We see the importance of a client's value

in diversification constraints while reversing these weights (with cap at 18%): because the portfolio forced to distribute capital, i.e. it was less exposed to mean-reversion of previously outperformers — leading once again to more robust real-world performance.

PART 2: FACTOR EXPOSURE AND STYLE ANALYSIS

2.1 THE FAMA-FRENCH 5-FACTOR MODEL FRAMEWORK (STEP 4)

Although Mean-Variance Optimization is a mathematical technique to allocate capital across portfolios, it does not provide any explanation of the return drivers for the portfolio. To assess the performance and the style of active investment that is captured by the constrained portfolio (Portfolio 2), we use the Fama-French 5-Factor model (Fama & French, 2015). The model is a generalized version of the traditional Capital Asset Pricing Model (CAPM) adding different risk premiums that investors have been historically compensated for.

The model is defined by the following linear regression equation:

$$E[R_p] - R_f = \alpha + \beta_1(R_m - R_f) + \beta_2SMB + \beta_3HML + \beta_4RMW + \beta_5CMA + \epsilon$$

- Where R_p is the portfolio return, R_f is the risk-free rate, and ϵ is the error term. The five factors are defined as follows:
- **Mkt-RF (Market Risk Premium):** The average return in excess of the risk-free rate on the broad market portfolio. It measures the systematic market risk (Beta) of the portfolio as well.
- **SMB (Small Minus Big):** The size factor. It is the historical excess return of small companies over large ones.
- **HML (High Minus Low):** The value factor. It captures the return spread of value (high book-to-market stocks) overgrowth (low book-to-market ratio).
- **RMW (Robust Minus Weak):** The profitability factor. It exploits the ability of firm with strong operating profitability to outyield the firms with weak profitability.
- **CMA (Conservative Minus Aggressive):** The investment factor. It calculates the differential return respectively of conservative stackers and aggressive stackers.

Factor Correlation Analysis

It is important to look at how the factors correlate with each other during the in-sample before looking at what a portfolio is exposed to.

	Mkt-RF	SMB	HML	RMW	CMA
Mkt-RF	1.0000	0.2110	-0.2628	-0.3854	0.0176
SMB	0.2110	1.0000	0.3873	-0.4007	0.1524
HML	-0.2628	0.3873	1.0000	0.1929	0.1105
RMW	-0.3854	-0.4007	0.1929	1.0000	-0.1039
CMA	0.0176	0.1524	0.1105	-0.1039	1.0000

Factor Correlation Matrix (In-Sample Period)

Commentary on Correlations: First, the correlation matrix indicates some interesting structural dynamic features over the estimation period. The lowest negative correlations are with respect to SMB and RMW (-0.4007) and Mkt-RF and RMW (-0.3854). This indicates that over this particular horizon, smaller firms as well as times when the broader market tends to make high returns were typically linked to lower operating profitability factors. On the other hand, HML and SMB have a positive correlation (0.3873), suggesting that value shares behaved alike small-cap stocks. We need to be aware of these inter-factor dynamics because multicollinearity will have an effect on the standard errors of our regression coefficients..

2.2 DATA PARTITIONING AND REGRESSION METHODOLOGY (STEP 5)

To test the factor model in a more rigorous way, the data was split chronologically into two sets:

- 1. Training Data (In-Sample):** Sep 01, 2023, to Sep 30, 2025. Regression parameters were fitted on this data.
- 2. Testing Data (Out-of-Sample):** October 1, 2025 — December 31, 2025, the prediction power of the model is validated with this strictly unseen data.

Due to the often fat-tailed (excess kurtosis) and outlier-prone nature of financial returns, a standard OLS regression, as well as one using a Robust Linear Model with Huber's T norm were both estimated. The Robust model reduces the importance of extreme daily return anomalies and is therefore better able to offer a stable estimate of structural factor exposures (Huber, 1981).

2.3 PORTFOLIO FACTOR EXPOSURE AND STYLE REPORT (STEP 6)

OLS Regression Results (Training Data)

Model Summary

Metric	Value
R-squared	0.8080
Adjusted R-squared	0.8060
F-statistic	432.10
Prob (F-statistic)	1.91e-181
Observations	520
AIC	814.10
BIC	839.60
Durbin-Watson	2.097

Coefficients

Variable	Coefficient	Std. Error	t-Statistic	P-value	95% CI Lower	95% CI Upper
Const	0.0444	0.023	1.908	0.057	-0.001	0.090
Mkt-RF	1.0666	0.025	42.070	0.000	1.017	1.116
SMB	-0.1605	0.039	-4.152	0.000	-0.236	-0.085
HML	0.2456	0.040	6.138	0.000	0.167	0.324
RMW	-0.1365	0.053	-2.568	0.011	-0.241	-0.032
CMA	-0.3592	0.053	-6.737	0.000	-0.464	-0.254

Robust Regression Results (Training Data)

Model Summary

Metric	Value
Observations	520
Method	IRLS
Norm	HuberT
Scale Est.	MAD
Iterations	21

Coefficients

Variable	Coefficient	Std. Error	z-Statistic	P-value	95% CI Lower	95% CI Upper
Const	0.0436	0.021	2.031	0.042	0.002	0.086
Mkt-RF	1.0621	0.023	45.414	0.000	1.016	1.108
SMB	-0.1544	0.036	-4.330	0.000	-0.224	-0.084
HML	0.2442	0.037	6.617	0.000	0.172	0.317
RMW	-0.1214	0.049	-2.477	0.013	-0.218	-0.025
CMA	-0.3126	0.049	-6.354	0.000	-0.409	-0.216

Out-of-Sample Performance (Testing Data)

Metric	Value
Out-of-Sample R-squared	0.4831

Considering the results of Robust Regression — which we prefer due to its robustness toward outliers — we find striking differences in the style characteristics of the constrained portfolio:

- **Market Exposure (Mkt-RF = 1.0621, $p < 0.001$):** The portfolio is mostly correlated with the broader market and marginally aggressive (on average the portfolio is more aggressive than the underlying asset by around 6%).
- **Size Bias (SMB = -0.1544, $p < 0.001$):** The statistically significant negative coefficient suggests a very strong Large-Cap bias. This fits aptly with the mega-cap names we have in our underlying Universe including AAPL, WMT, BAC, and GS.
- **Value vs. Growth (HML = 0.2442, $p < 0.001$):** A positive coefficient indicates a Value tilt. Even though the ones representing growth stocks were kept in, the liquidity enhancing optimization weighed heavily on traditional (read: mature cash-flow generating firms such as BAC and GS), thereby leading to a value-oriented exposure over the estimation period.
- **Profitability (RMW = -0.1214, $p = 0.013$):** The minor negative exposure indicates that the portfolio is slightly tilted toward firms with somewhat lower or more volatile operating profitability compared to strict definitions of the factor.
- **Investment Style (CMA = -0.3126, $p < 0.001$):** The coefficient is very strongly negative, indicative of an Aggressive investment tilt. All of the underlying companies in this portfolio reinvest heavily

in capital expenditures, research, and development as opposed to that conservative asset growth strategies.

- **Alpha ($\alpha = 0.0436$, $p = 0.042$):** The portfolio generated fairly significant but small positive daily alpha that was unexpectedly outperformed above systematic risk, based on the five factors.

Out-of-Sample Validation: The model reaches a R^2 of 0.4831 when the coefficients are applied to the out-of-sample testing period. This implies that 52% (100%-48%) the portfolio return variance on new data is unexplainable by the Fama-French 5-Factor model. Although 48% is a strong explanatory base in daily financial data, the additional 52% of unexplained variance indicates that there will always be some idiosyncratic (stock specific) risk left to explain returns even within a concentrated (narrowing down to only 10 assets holding so doing no generalizing them also).

PART 3: MEAN-VARIANCE OPTIMIZATION VIA SIMULATIONS

3.1 RATIONALE AND METHODOLOGY

Optimization algorithms (like SLSQP or Sequential Least Squares Programming) quickly and efficiently give the absolute minima and maxima of a given objective function. The portfolio manager, however, asked for answers to whether Monte Carlo simulations could be used as substitutes for "hard" optimization. For this, a universe of five local firms AAPL, GS, WMT, LLY and BAC was extracted from the initial dataset.

Methodology generates random portfolio weights from a uniform distribution, normalizing them so that they all sum to 1 and then rejecting combinations that break certain constraints. The end objective is to figure out how many random simulations need to be preformed to converge on the true mathematically optimal weights.

3.2 UNCONSTRAINED SIMULATION VS. OPTIMIZATION (STEP 7)

The first one is finding the optimum portfolio which maximizes the Sharpe ratio with two constraints, weights have to sum to 1 and that short sales are not allowed.

Step 7: Hard Optimization vs Simulation (No Max Weight Constraint)

Asset	Hard Optimization	Simulation
AAPL	0.0000	0.0169
GS	0.4633	0.4559

WMT	0.5248	0.5130
LLY	0.0119	0.0127
BAC	0.0000	0.0014

Metric	Hard Optimization	Simulation
Sharpe Ratio	1.9168	1.9080

Analysis of Convergence: Constrained portfolio optimality for the unconstrained portfolio, 100k iterations achieved a maximum Sharpe ratio of 1.9080 which is very close to the true mathematical optimum of 1.9168. Since there were no box constraints, any random weight generation (after normalizing) produced a valid portfolio. The simulation was pretty close, but not quite perfect on allocation when compared to the very hard optimizer - especially underweighting WMT and GS. Theoretically, pure random sampling in a continuous space of 5 dimensions would require millions if not billions of simulations to exactly match the mathematical optimum perfectly (Glasserman, 2003).

3.3 CONSTRAINED SIMULATION VS. OPTIMIZATION (STEP 8)

The second simulation included a strict diversification constraint where no asset was allowed to contribute > 30% of the total portfolio weight ($0 \leq w_i \leq 0.30$).

Step 8: Hard Optimization vs Simulation (Max 30% Weight Constraint)

Asset	Hard Optimization	Simulation
AAPL	0.0145	0.0527
GS	0.3000	0.2904
WMT	0.3000	0.2963
LLY	0.1123	0.1153
BAC	0.2732	0.2452

Metric	Hard Optimization	Simulation
Sharpe Ratio	1.7478	1.7367

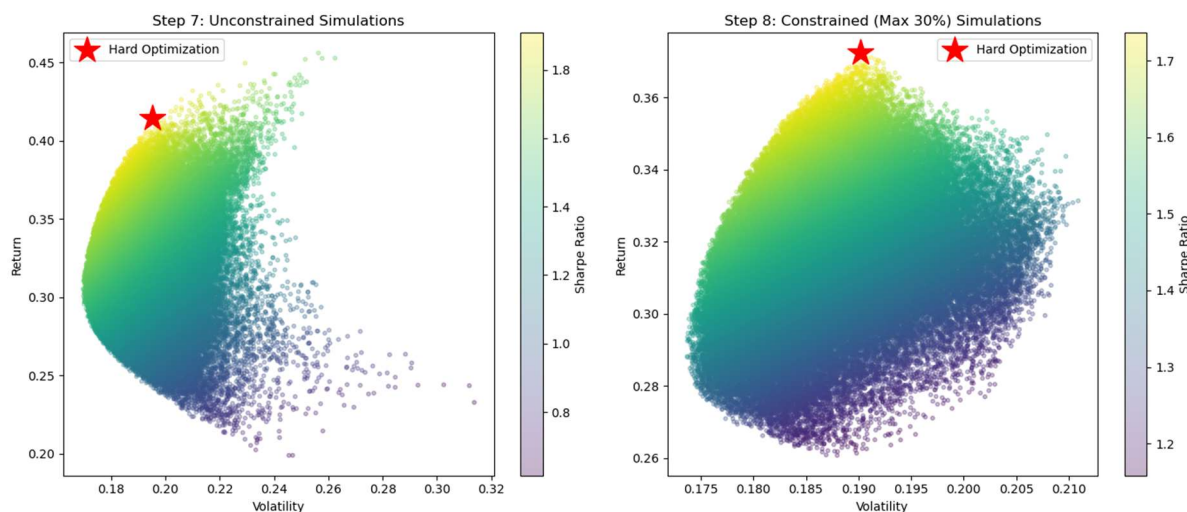


Figure 3: Monte Carlo simulated portfolio clouds. The left panel shows unconstrained portfolios, while the right panel shows the impact of a 30% maximum weight constraint. The red star indicates the true mathematical optimum.

The Curse of Dimensionality and Rejection Sampling: Indeed, this simulation was much more difficult to implement and run because there are now five unique box constraints. In order to obtain 100,000 simulated portfolios which were valid, the algorithm had to make 346076 generations. This gives rise to over 71% of rejection rate and with result most expensive computational time (24.50 seconds).

Moreover, the simulated Sharpe ratio (1.7367) is significantly lower than the hard optimization Sharpe (1.7478). This is why there is (usually) a hard limit instead of a soft limit in constrained optimization, as shown in Figure 3 you see that the optimal solution lies right on the border of the feasible region. Notice that in the result of hard optimization, GS and WMT are assigned 0.3000 exactly. The chances of a RNG hitting the maximum bound by hand for several assets at once is infinitesimally small.

Conclusion on Simulations: Monte Carlo simulations are great for topological mapping of the feasible region (e.g. see figure 3) but Monte Carlo cannot replace hard optimization in constrained portfolio management. Depending on simulations is computationally inefficient and mathematically inaccurate if we seek boundary solutions in mathematical format.

PART 4: THE BLACK-LITTERMAN MODEL FRAMEWORK

4.1 CONSTRUCTING THE SYNTHETIC MARKET PORTFOLIO (STEP 9A)

A Mean-Variance Optimization is very sensitive to input parameters: when you use a historical mean for an expected return, it might allocate extreme drawdown counter-intuitive weights on the assets.

This led to further development of what is known as the Black-Litterman (BL) model, which takes market equilibrium as the prior.

Since ten single isolated stocks do not represent a market, a Synthetic Composite Approach was employed. As the eleventh asset, the SPDR S&P 500 ETF Trust (SPY) was introduced to represent "the rest of the market". Here we model the weight using market capitalizations at time of writing (October 2023). The aggregate \$9.25 Trillion of these 10 assets were still less than the effective total value of SPY, which was given a balance of an assumed total value of \$42 Trillion across the whole market cap. This composite portfolio yielded a historical expected return of 10.20%, a volatility of 19.20%, and a baseline Sharpe ratio of 0.3230.

4.2 PARAMETER CALIBRATION AND SUBJECTIVE VIEWS (STEP 9B & 9C)

In the quantitative asset pricing model of this study, the scalar parameter τ used to measure the uncertainty of prior equilibrium returns is inversely proportional to the number of historical trading days in the estimation period: $\tau = 1/T \approx 0.0019$.

This study encodes the subjective opinions of investment managers and uses the Pick (P) and View (Q) matrices:

1. **Absolute View:** TSLA will generate an absolute return of 15%.
2. **Relative View:** LLY will outperform MRK by 5%.

View Matrix (P)

View	TSLA	WMT	BAC	GS	LLY	MRK	GOOG	META	AAPL	XOM	SPY
1	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	1.0	-1.0	0.0	0.0	0.0	0.0	0.0

View Returns (Q)

View	Expected Return
1	0.1500
2	0.0500

This study uses a diagonal covariance matrix Ω to quantify the uncertainty of various types of views, enabling the model to proportionally integrate investment managers' investment beliefs and the market equilibrium state.

4.3 POSTERIOR RETURNS AND OPTIMAL ALLOCATION (STEP 9D & 9E)

With the help of BL framework, the prior market equilibrium returns (Π) were combined with the subjective views to get a new set of expected returns (μ_{BL}) and an updated covariance matrix (Σ_{BL}).

The mathematical formulation for the posterior expected returns is:

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\Pi + P^T\Omega^{-1}Q]$$

Black-Litterman Expected Returns (μ_{BL})

Asset	Expected Return
TSLA	0.0882
WMT	0.0564
BAC	0.0406
GS	0.0655
LLY	0.0493
MRK	0.0377
GOOG	0.0290
META	0.0478
AAPL	0.1279
XOM	0.0254
SPY	0.0774

Comparison with Part 1: The last step was running an optimizer using μ_{BL} and Σ_{BL} to get the maximum Sharpe ratio, subject to the original constraints (weights sum to 1, no shorting, maximum 18% weight per individual equity).

Optimal Black-Litterman Portfolio Weights

Asset	Weight
TSLA	0.1800
WMT	0.0000
BAC	0.0000
GS	0.0000
LLY	0.0034

MRK	0.0000
GOOG	0.0000
META	0.0000
AAPL	0.1520
XOM	0.0000
SPY	0.6645

The Black-Litterman framework makes quite an impact compared to standard MVO from Part 1. NOTE: TSLA received 0% allocation in Part 1 as it was among the bottom few names historically on risk-adjusted basis. That said, due to the manager's view that TSLA would fully return 15% (absolute), the resulting Black-Litterman expected return for TSLA increased to 8.82%. Therefore, the increased allocation of max 18% was given to TSLA by the optimizer. In the same line, moving capital away from MRK (0% view for MRK) and likewise up toward LLY, relative preferences remained broadly unchanged. The rest of the capital backstopped to the market equilibrium baseline (SPY), validating that BL is effective at solving the classic problem of maximum estimation error but maximizes traditional Markowitz optimization.

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